

ESIL', Kazakhstan

WMO: 350680

Lat: 51.9733N

Lon: 66.4183E

Elev: 725

StdP: 14.31

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-20.7	-31.3	1.0	-25.5	-26.2	1.3	-20.5	27.0	15.5	22.4	12.9	2.4	50	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	91.0	64.3	87.8	63.7	84.5	62.9	67.9	82.6	66.4	81.7	65.1	79.9	7.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	89.0	72.9	61.4	83.6	71.8	59.5	78.1	70.8	32.7	82.4	31.5	81.7	30.4	80.0	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	21.0	18.3	DB	-31.8	96.6	4.5	3.7	-35.0	99.3	-37.6	101.5	-40.1	103.6	-43.4	106.3	(4)
(5)			WB	-31.8	71.0	4.3	2.2	-34.9	72.5	-37.4	73.8	-39.8	75.0	-42.9	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	2.8	5.8	19.8	42.8	57.3	66.2	68.2	66.1	54.7	40.7	22.5	8.9		
	DBStd	26.13	13.19	12.30	11.84	11.05	8.71	7.65	6.39	7.86	8.64	9.11	13.44	13.12		
	HDD50	6399	1464	1239	937	264	32	2	0	1	48	311	826	1275		
	HDD65	10217	1929	1659	1402	668	264	76	41	83	326	753	1276	1740		
	CDD50	2065	0	0	0	47	257	486	564	500	188	23	0	0		
	CDD65	408	0	0	0	0	24	111	140	117	16	0	0	0		
	CDH74	5268	0	0	0	20	440	1447	1627	1471	257	6	0	0		
	CDH80	1895	0	0	0	2	108	543	580	580	82	0	0	0		
	WSAvg	7.0	6.2	7.3	7.7	7.6	7.6	7.0	6.7	6.1	6.3	6.6	7.0	7.4		
	PrecAvg	11.5	0.6	0.5	0.6	0.7	1.3	1.4	2.0	1.1	0.6	0.8	0.9	0.8		
	PrecMax	18.3	1.3	1.2	1.5	1.7	3.1	5.4	4.1	2.8	1.3	2.3	1.5	1.9		
(15) Precipitation	PrecMin	6.1	0.1	0.0	0.0	0.0	0.1	0.2	0.4	0.1	0.1	0.0	0.3	0.2		
	PrecStd	3.1	0.3	0.3	0.4	0.4	0.8	1.2	1.1	0.8	0.4	0.5	0.4	0.4		
	DB	33.2	33.4	50.5	76.9	87.9	93.5	94.6	94.9	88.9	73.4	50.4	34.2			
	MCWB	32.3	31.8	43.4	55.7	61.0	65.1	65.5	65.2	62.0	52.4	43.8	32.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.0	31.1	41.1	70.8	82.8	89.7	90.0	90.5	81.0	66.3	44.9	32.2			
	2% MCWB	28.9	30.0	37.6	53.1	59.6	63.8	65.2	64.4	57.8	50.3	41.2	31.0			
	5% DB	26.7	28.2	37.5	65.8	78.9	86.5	86.1	86.5	75.5	60.8	40.9	29.7			
	5% MCWB	25.6	27.1	34.9	51.4	58.0	62.9	65.1	63.8	56.6	47.7	37.8	28.5			
	10% DB	23.2	24.6	34.9	60.8	75.0	82.4	82.7	82.3	70.6	55.6	37.6	26.5			
	10% MCWB	22.2	23.5	33.0	49.5	56.4	62.1	64.0	62.6	54.4	45.7	35.1	25.2			
	0.4% WB	32.5	32.2	43.4	58.3	63.9	68.7	70.2	68.8	63.4	54.9	45.8	32.9			
	0.4% MCDB	33.0	33.2	49.7	69.9	79.5	84.4	83.3	84.9	82.5	69.2	48.2	33.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	29.0	30.0	37.8	55.1	61.4	66.8	68.6	66.7	59.8	51.6	41.4	31.1			
	2% MCDB	29.9	31.1	40.6	67.3	77.8	82.6	82.1	83.8	76.0	62.9	44.7	32.0			
	5% WB	25.7	27.1	35.1	52.5	59.5	65.2	67.0	65.4	57.5	48.8	38.1	28.4			
	5% MCDB	26.6	28.1	37.1	63.7	74.8	81.1	80.8	81.8	71.8	58.5	40.6	29.6			
	10% WB	22.2	23.5	33.2	49.8	57.7	63.5	65.6	63.9	55.6	46.4	35.2	25.2			
	10% MCDB	23.3	24.5	35.0	60.4	72.1	79.0	79.1	79.1	68.8	54.7	37.5	26.4			
	MDBR	14.8	15.9	15.5	19.9	23.8	22.9	21.7	23.2	22.5	18.0	13.0	14.3			
	5% DB	13.5	12.1	13.4	29.0	30.3	28.9	27.7	29.1	29.8	28.6	14.7	12.8			
(36) Mean Daily Temperature Range	MCWBR	13.3	11.7	10.6	14.8	13.0	11.1	10.1	10.8	13.1	15.7	11.5	12.2			
	MCDBR	13.4	12.1	12.9	25.9	25.9	24.5	22.7	25.8	26.0	23.9	13.4	12.5			
	MCWBR	13.3	11.9	10.6	14.7	12.7	10.9	9.6	10.8	12.5	14.3	11.2	12.2			
	MCWBR	13.3	11.9	10.6	14.7	12.7	10.9	9.6	10.8	12.5	14.3	11.2	12.2			
(40) Clear-Sky Solar Irradiance	taub	0.293	0.325	0.378	0.411	0.408	0.407	0.421	0.420	0.367	0.349	0.327	0.293			
	taud	2.060	2.008	1.961	2.118	2.212	2.287	2.248	2.225	2.324	2.277	2.149	2.057			
	Ebn at Noon	213	240	250	260	268	269	262	255	257	234	198	187			
	Edh at Noon	25	36	46	44	43	40	41	40	32	27	23	21			
(44) All-Sky Solar Radiation	RadAvg	421	781	1225	1512	1887	1997	1867	1602	1180	644	301	278			
	RadStd	31	43	131	151	146	148	135	137	87	85	33	41			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page